

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	183210	49.681238
1	[Git, Change, Create]	53699	14.561611
2	[Git, Change, Delete]	28358	7.689867
3	[Java, ByteCode, Method, Member]	13477	3.654572
4	[Java, ByteCode, Parameter]	13293	3.604676
5	[Git, Commit]	13218	3.584338
6	[File, Git]	11288	3.060978
7	[Git, Change, Rename]	10989	2.979898
8	[Java, ByteCode, Bound, ParameterizedType]	7279	1.973854
9	[Java, ByteCode, Bound]	7240	1.963278
10	[Java, ByteCode, Member, Field]	3605	0.977571
11	[Java, ByteCode, Bound, WildcardType]	2947	0.799141
12	[Java, Value, ByteCode, Annotation]	2931	0.794802
13	[Xml, Element]	2162	0.586272
14	[Java, ByteCode, Constructor, Method, Member]	2118	0.574340
15	[Xml, Text]	1450	0.393198
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.301271
17	[Java, ByteCode, Method, Member, Lambda]	968	0.262494
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.240800
19	[Type, File, Java, ByteCode, Class]	846	0.229411
20	[Json, Key]	702	0.190362
21	[Value, Json, Scalar]	685	0.185752
22	[Java, Value, ByteCode, Primitive]	672	0.182227
23	[Type, File, Java, ByteCode, JavaType]	649	0.175990
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.156737
25	[Type, File, Java, ByteCode, ExternalType]	411	0.111451
26	[Git, Change, Copy]	342	0.092740
27	[Author, Git, Person]	299	0.081080
28	[Value, Array]	287	0.077826
29	[Committer, Git, Person]	252	0.068335

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

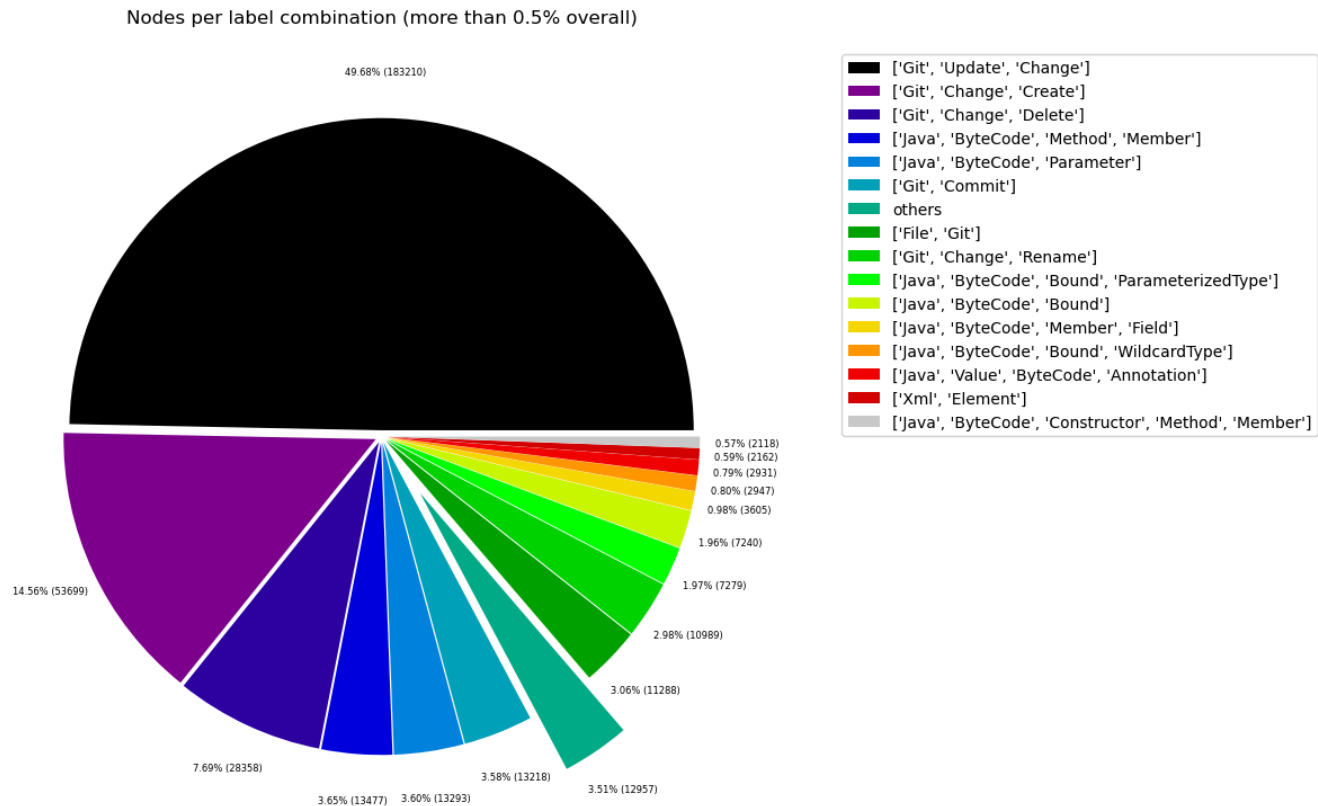


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000271
1	[Package, File, Json, NPM]	1	0.000271
2	[Repository, File, Git]	1	0.000271
3	[File, TS, Scan]	1	0.000271
4	[File, Json]	2	0.000542
5	[File]	3	0.000814
6	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001085
7	[Maven, Exclusion]	5	0.001356
8	[Value, Array, Json]	6	0.001627
9	[Dependency, NPM]	7	0.001898
10	[Type, File, Java, ByteCode, Void]	9	0.002441
11	[File, Maven, Xml, Pom, Document]	9	0.002441
12	[Java, ManifestSection]	9	0.002441
13	[File, Java, Manifest]	9	0.002441
14	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002441
15	[File, Java, ServiceLoader]	10	0.002712
16	[File, Java, Properties]	12	0.003254
17	[Maven, ExecutionGoal]	16	0.004339
18	[Maven, PluginExecution]	16	0.004339
19	[Xml, Attribute]	18	0.004881
20	[jQAssistant, Rule, Concept]	19	0.005152
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005695
22	[Maven, Configuration]	21	0.005695
23	[Maven, Plugin]	21	0.005695
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.005966
25	[Git, Branch]	28	0.007593
26	[Type, File, Java, ByteCode, Enum]	28	0.007593
27	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008135
28	[Xml, Namespace]	36	0.009762
29	[Type, File, Java, ByteCode, Annotation]	44	0.011932

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

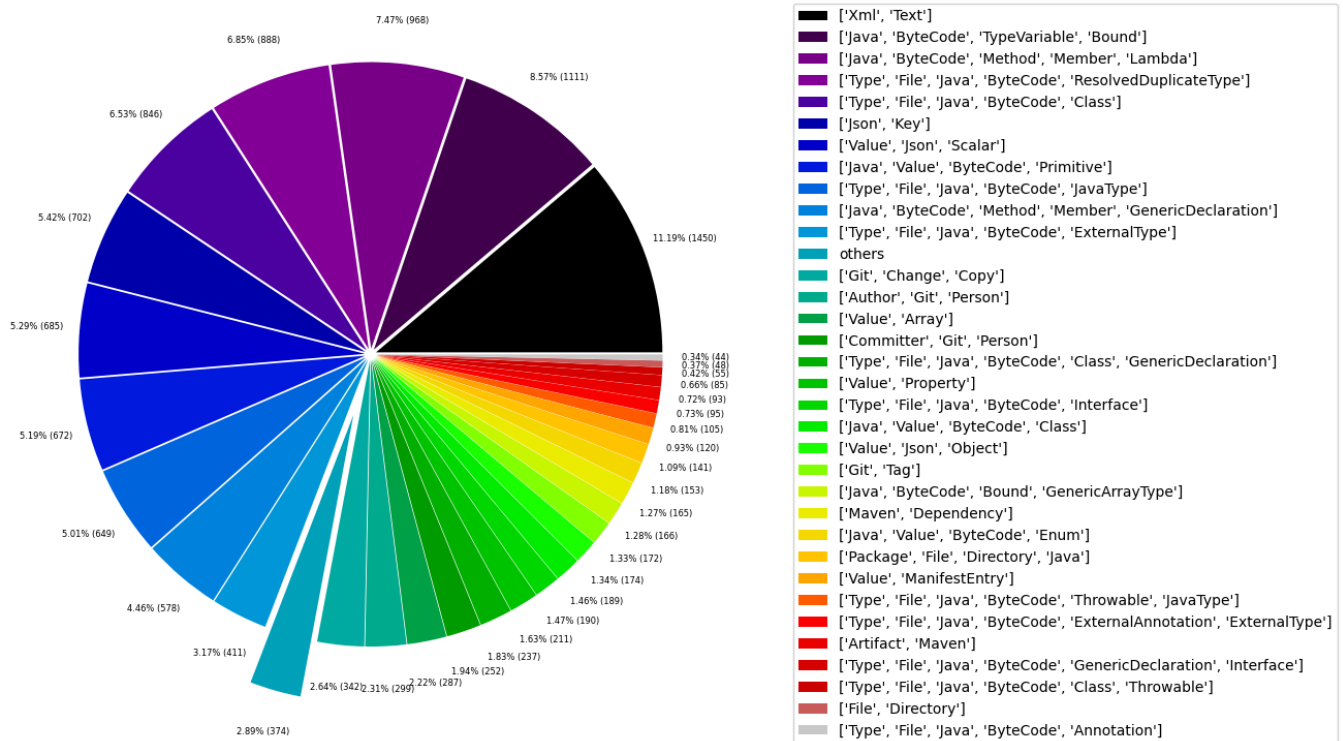


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

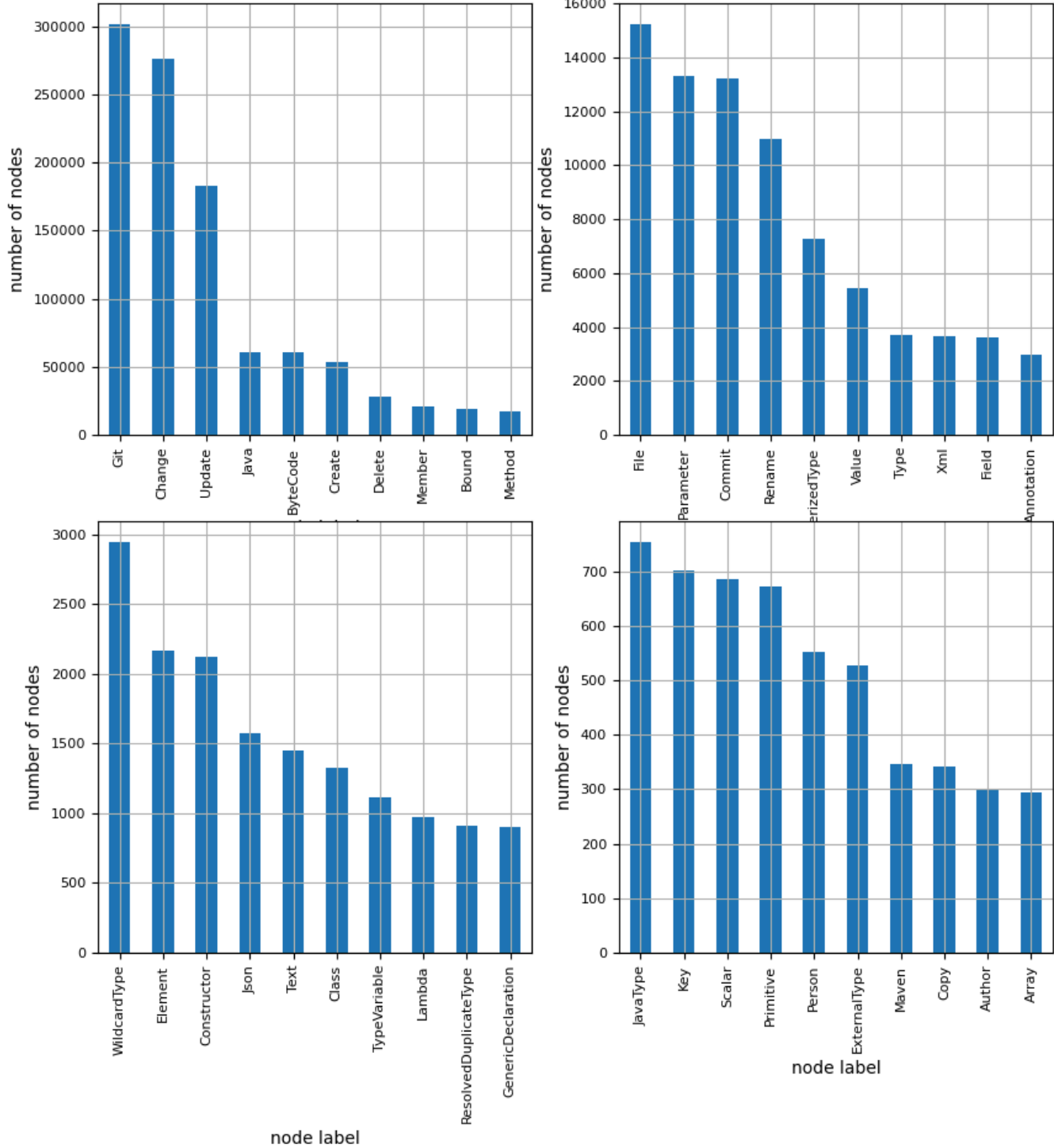
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	301856	81.854593
1	Change	276598	75.005356
2	Update	183210	49.681238
3	Java	60636	16.442725
4	ByteCode	60446	16.391202
5	Create	53699	14.561611
6	Delete	28358	7.689867
7	Member	20750	5.626798
8	Bound	18743	5.082558
9	Method	17145	4.649227
10	File	15249	4.135087
11	Parameter	13293	3.604676
12	Commit	13218	3.584338
13	Rename	10989	2.979898
14	ParameterizedType	7279	1.973854
15	Value	5428	1.471916
16	Type	3715	1.007400
17	Xml	3675	0.996553
18	Field	3605	0.977571
19	Annotation	2975	0.806734
20	WildcardType	2947	0.799141
21	Element	2162	0.586272
22	Constructor	2122	0.575425
23	Json	1570	0.425738
24	Text	1450	0.393198
25	Class	1327	0.359844
26	TypeVariable	1111	0.301271
27	Lambda	968	0.262494
28	ResolvedDuplicateType	910	0.246766
29	GenericDeclaration	904	0.245139
30	JavaType	754	0.204463
31	Key	702	0.190362
32	Scalar	685	0.185752
33	Primitive	672	0.182227
34	Person	551	0.149415
35	ExternalType	527	0.142907
36	Maven	346	0.093825
37	Copy	342	0.092740
38	Author	299	0.081080
39	Array	293	0.079453

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1144320

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	276598	24.171386
1	MODIFIES	276598	24.171386
2	UPDATES	183210	16.010382
3	CREATES	65030	5.682851
4	DELETES	39347	3.438461
5	INVOKES	36545	3.193600
6	COMMITTED	26436	2.310193
7	DEPENDS_ON	22492	1.965534
8	OF_TYPE	21920	1.915548
9	DECLARES	21211	1.853590
10	OF_RAW_TYPE	17353	1.516446
11	HAS_PARENT	16050	1.402580
12	HAS	14429	1.260924
13	HAS_COMMIT	13218	1.155096
14	RETURNS	12844	1.122413
15	HAS_FILE	11288	0.986437
16	RENAMES	10989	0.960308
17	READS	9379	0.819613
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.734672
19	HAS_NEW_NAME	6382	0.557711
20	OF_GENERIC_TYPE	5996	0.523979
21	RESOLVES_TO	5260	0.459662
22	SIMILAR	4078	0.356369
23	WRITES	3944	0.344659
24	CONTAINS	3936	0.343960
25	RETURNS_GENERIC	3593	0.313986
26	ANNOTATED_BY	2919	0.255086
27	REQUIRES	2230	0.194876
28	HAS_FIRST_CHILD	2162	0.188933
29	HAS_LAST_CHILD	2162	0.188933

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

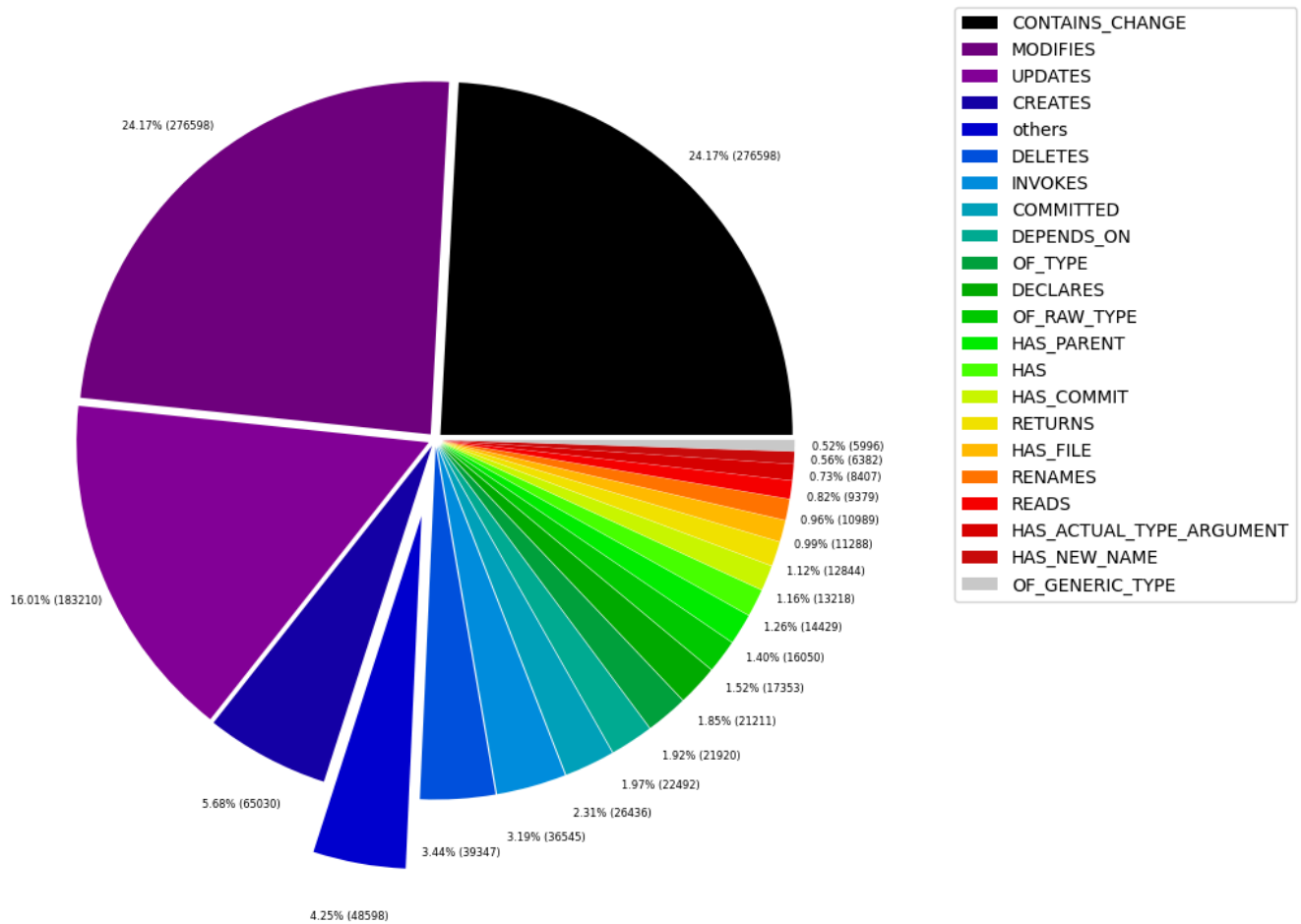


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

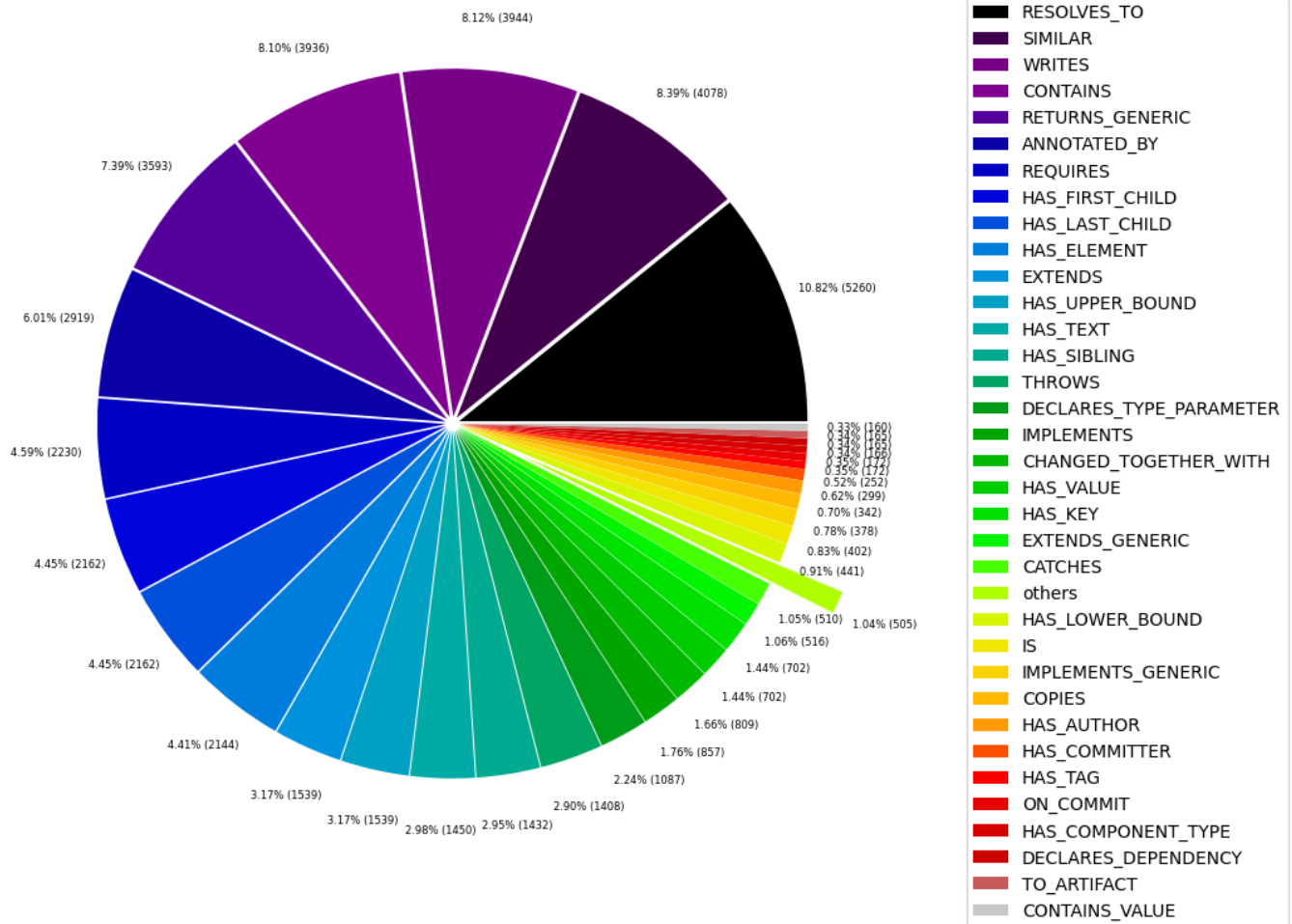
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000087
1	THROWS_GENERIC	5	0.000437
2	EXCLUDES	5	0.000437
3	DECLARES_DEV_DEPENDENCY	7	0.000612
4	DESCRIBES	9	0.000786
5	HAS_ROOT_ELEMENT	12	0.001049
6	HAS_GOAL	16	0.001398
7	HAS_EXECUTION	16	0.001398
8	OF_NAMESPACE	18	0.001573
9	HAS_ATTRIBUTE	18	0.001573
10	INCLUDES_CONCEPT	19	0.001660
11	IS_ARTIFACT	21	0.001835
12	HAS_CONFIGURATION	21	0.001835
13	USES_PLUGIN	21	0.001835
14	REQUIRES_TYPE_PARAMETER	24	0.002097
15	REQUIRES_CONCEPT	28	0.002447
16	HAS_BRANCH	28	0.002447
17	HAS_HEAD	29	0.002534
18	DECLARES_NAMESPACE	36	0.003146
19	HAS_DEFAULT	39	0.003408
20	COPY_OF	132	0.011535
21	CONTAINS_VALUE	160	0.013982
22	DECLARES_DEPENDENCY	165	0.014419
23	TO_ARTIFACT	165	0.014419
24	HAS_COMPONENT_TYPE	166	0.014506
25	ON_COMMIT	172	0.015031
26	HAS_TAG	172	0.015031
27	HAS_COMMITTER	252	0.022022
28	HAS_AUTHOR	299	0.026129
29	COPIES	342	0.029887

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	MODIFIES	[File, Git]	183210	183210	
1	[Git, Update, Change]	UPDATES	[File, Git]	183210	183210	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	183210	13218	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	53699	13218	
4	[Git, Change, Create]	MODIFIES	[File, Git]	53699	53699	
5	[Git, Change, Create]	CREATES	[File, Git]	53699	53699	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	28358	13218	
7	[Git, Change, Delete]	DELETES	[File, Git]	28358	28358	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	28358	28358	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22238	13377	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	16041	13218	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13218	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	13218	299	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13218	252	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11288	1	
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10989	13218	
16	[Git, Change, Rename]	DELETES	[File, Git]	10989	10989	
17	[Git, Change, Rename]	CREATES	[File, Git]	10989	10989	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	10989	10989	
19	[Git, Change, Rename]	RENAMES	[File, Git]	10989	10989	
20	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8506	13377	
21	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8444	13377	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6382	11288	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2519	2118	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 368771
total_number_of_relationships (edges): 1144320
-> total directed graph density: 8.4146323212388e-06
-> total directed graph density in percent: 0.00084146323212388
```